F12 4th

Lecture 20: Section 3.1
Exponential Functions

Exponential functions

Properties of exponential functions

Graphing exponential functions

One-to-one property of exponential functions

Natural base e and the natural exponential function

Compound interest

Continuous compound interest

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#7 change back to $e^{-x+3}-2$

Lecture 20: Section 3.1
Exponential Functions

<u>Def.</u> An **exponential function with base** a is a function of the form $f(x) = a^x$

where x is a real number, $a > 0$ and $a \neq 1$.

<u>ex.</u> Let $f(x) = 4^x$. Find the following:

1) $f(-2) = 4^{-2} = \frac{1}{4^2} = \boxed{\frac{1}{16}}$

2) $f\left(\frac{3}{2}\right) = 4^{3/2} = (4^{1/2})^3 = (\sqrt{4})^3 = 2^3 = \boxed{8}$

$(4^3)^{1/2} = \sqrt{64} = 8$

The domain of $f(x) = a^x$:

$(-\infty, \infty)$

NOTE: Be sure to see the difference between the **power function**

$$f(x) = x^4$$

and the **exponential function**

$$f(x) = 4^x$$

Can 2^x be negative? No

Graphs of Exponential Functions

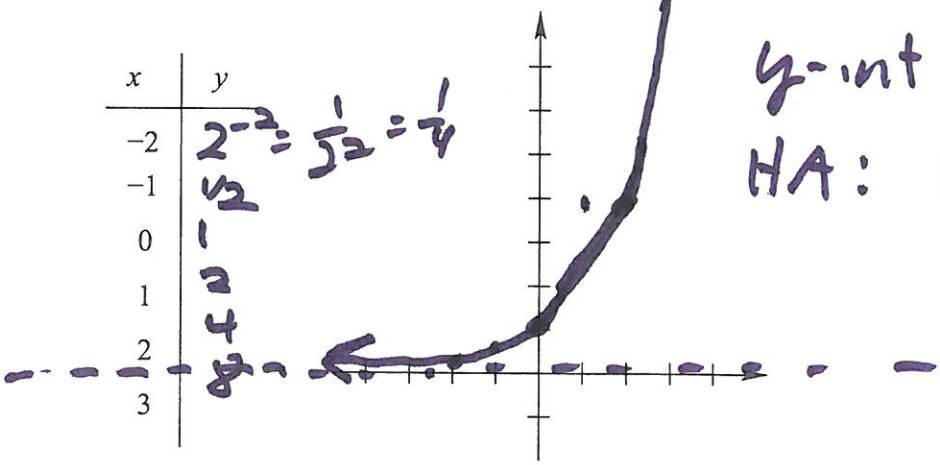
<u>ex.</u> $f(x) = 2^x$

*Note: $a > 1$

y-int $(0,1)$

HA: $y=0$

x	y
-2	$2^{-2} = \frac{1}{2^2} = \frac{1}{4}$
-1	$1/2$
0	1
1	2
2	4
3	8



memorize basic shape.

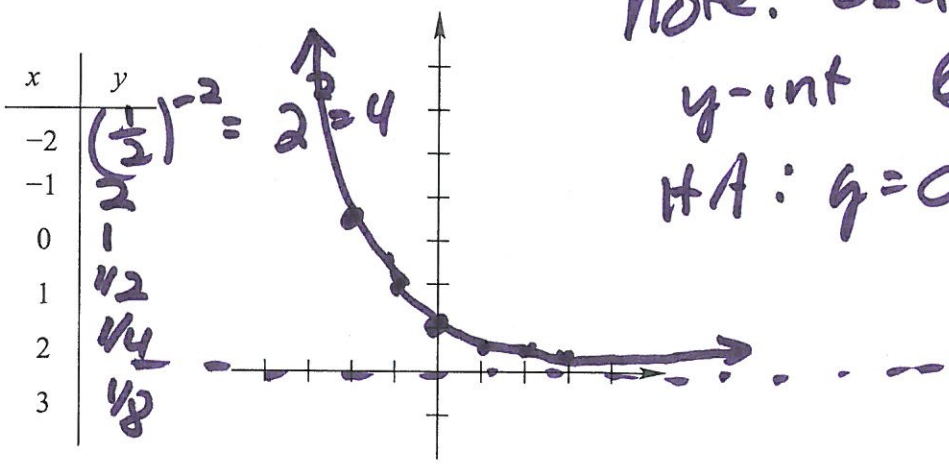
<u>ex.</u> $g(x) = \left(\frac{1}{2}\right)^x$

*Note: $0 < a < 1$

y-int $(0,1)$

HA: $y=0$

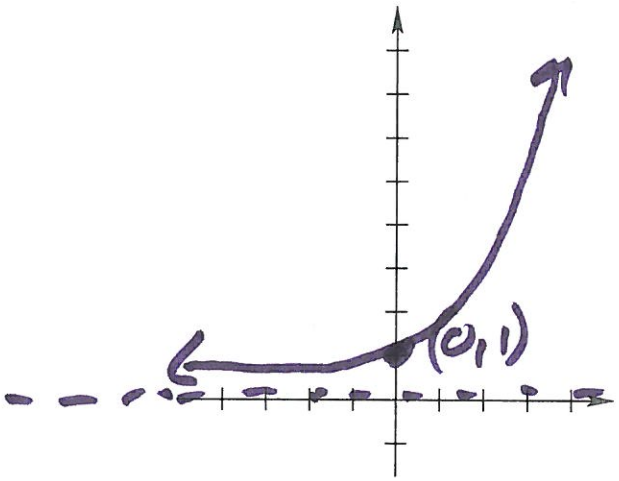
x	y
-2	$\left(\frac{1}{2}\right)^{-2} = 2^2 = 4$
-1	2
0	1
1	$1/2$
2	$1/4$
3	$1/8$

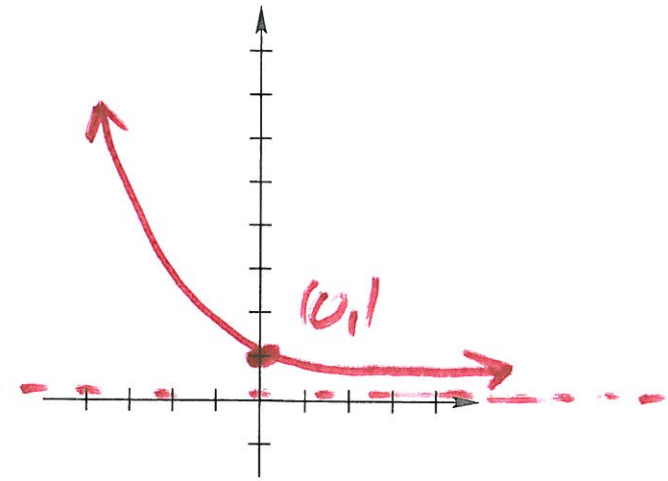


NOTE: The graph of $g(x)$ is the reflection of f across the y-axis since

$$g(x) = \left(\frac{1}{2}\right)^x = (2^{-1})^x = 2^{-x} = f(-x)$$

Properties of Exponential Functions





$f(x) = a^x, a > 1$

1. Domain: $(-\infty, \infty)$
2. Range: $(0, \infty)$
3. Intercept(s): y-int $(0,1)$ no x-int.
4. Asymptote: HA: $y=0$
5. $f(x)$ is increasing and one-to-one $\therefore f^{-1}$ exists
6. Points $\left(-1, \frac{1}{a}\right)$, $(1, a)$ are on each graph

$f(-1) = a^{-1} = \frac{1}{a}$

$f(1) = a^1 = a$

$f(x) = a^x, 0 < a < 1$

(or $f(x) = a^{-x}, a > 1$)

1. Domain: $(-\infty, \infty)$
2. Range: $(0, \infty)$
3. Intercept(s): y-int $(0,1)$ no x-int.
4. Asymptote: HA: $y=0$
5. $f(x)$ is decreasing and one-to-one $\therefore f^{-1}$ exists

Using Translations to Graph

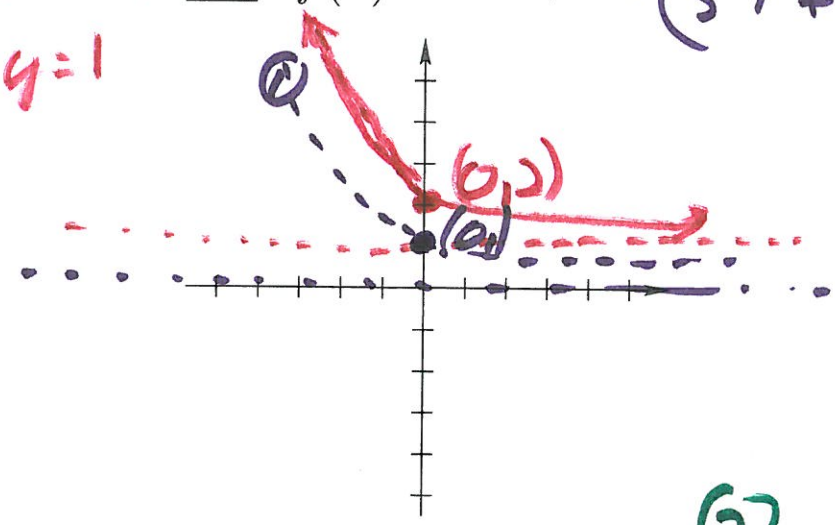
D: $(-\infty, \infty)$

R: $(1, \infty)$

HA: $y = 1$

Graph each function and find the domain, range and asymptote of the function:

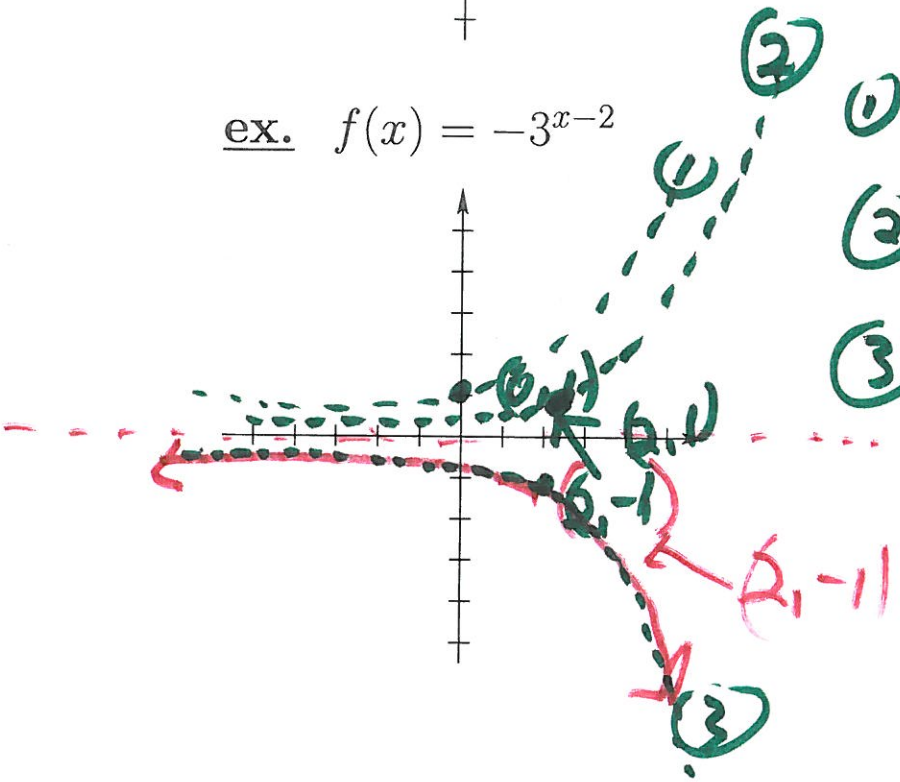
<u>ex.</u> $f(x) = 3^{-x} + 1 = (3^{-1})^x + 1 = \left(\frac{1}{3}\right)^x + 1$



① $y = \left(\frac{1}{3}\right)^x$

② $y = \left(\frac{1}{3}\right)^x + 1$ ↑

<u>ex.</u> $f(x) = -3^{x-2}$



① $y = 3^x$

② $y = 3^{x-2}$ → 2

③ $y = -3^{x-2}$ reflect across the x-axis

D: $(-\infty, \infty)$

R: $(-\infty, 0)$

HA: $y = 0$

One-to-one Property of Exponential Functions: For $a > 0$ and $a \neq 1$,

$$a^x = a^y \text{ if and only if } x = y$$

This allows us to solve exponential equations:

If variables are in the exponents, get the same base on both sides.

ex. Solve for x:

1) $8^{-x} = \left(\frac{1}{4}\right)^{x+3}$

$(2^3)^{-x} = (2^{-2})^{x+3}$

$2^{-3x} = 2^{-2x-6}$

thus $-3x = -2x-6$

$-2(x+3) = -2x-6$

$-x = -6$

$\boxed{x = 6}$

can check sol.

2) $9^x = \frac{3^{x^2}}{27}$

$(3^2)^x = \frac{3^{x^2}}{3^3}$

$3^{2x} = 3^{x^2} \cdot 3^{-3}$

$3^{2x} = 3^{x^2-3}$

thus $2x = x^2 - 3$

$x^2 - 2x - 3 = 0$

$(x-3)(x+1) = 0$

$\boxed{x = 3 \quad x = -1}$

can check sol.

Note: $4^{x^2} \overset{?}{\neq} (4^x)^2 = 4^{2x}$

$$x^2 \neq 2x$$

ie. let $x = 3$

$$4^{3^2} \overset{?}{\neq} (4^3)^2$$

$$4^9 \neq 4^6$$

The Natural Base e → e is an irrational

Consider the expression $\left(1+\frac{1}{n}\right)^n$

n	$\left(1+\frac{1}{n}\right)^n$
10	2.59374246
100	2.704813829
1000	2.716923932
10,000	2.718145927
100,000	2.718268237
1,000,000	2.718280469
$\downarrow$	$\downarrow$
∞	?

$2 < e < 3$

Def. $e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n \approx 2.718281828\ldots$

NOTE: e is an irrational number. since it never ends and never repeats.

$\cancel{2 < e^x < e^2}$ $\quad 2^x < e^x < 3^x$

Def. The function given by $f(x) = e^x$ is called the **natural exponential function** with natural base e.

ex. Find: $f(0) = e^0 = \boxed{1}$ $\quad f(-1) = e^{-1} = \boxed{\frac{1}{e}}$

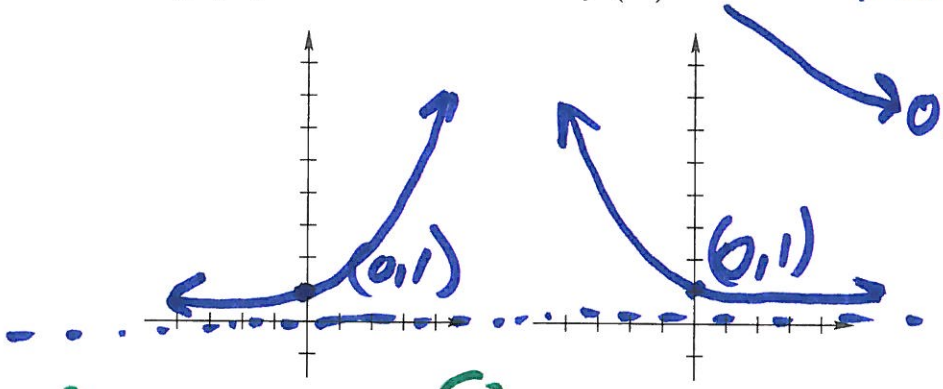
$f(1) = e^1 = \boxed{e}$ $\quad f(2) = \boxed{e^2}$

Sketch the graphs:

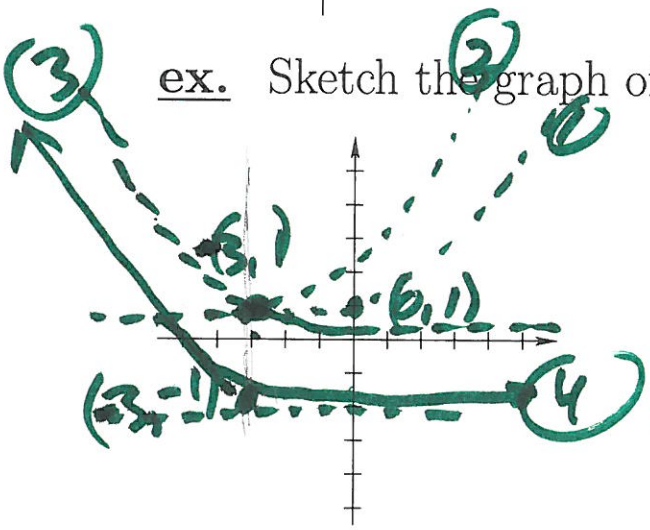
$f(x) = e^x$ $\qquad$ $f(x) = e^{-x}$ reflect. across the y-axis

or $e^{-x} = (e^{-1})^x = \left(\frac{1}{e}\right)^x$

base is between 0 and 1



ex. Sketch the graph of $f(x) = e^{-(x+3)} - 2$



$\cancel{y = e^{-x}}$

① $y = e^x$

② $y = e^{(x+3)}$ $\leftarrow 3$

③ $y = e^{-(x+3)}$ refl. across the y axis

④ $y = e^{-(x+3)} - 2$

Applications: Compound Interest

Suppose a principal P is invested at an annual interest rate r, compounded once a year. If the interest is added to the principal at the end of each year, then

Year	Balance after each compounding
0	$P_0 = P$
end of 1 yr	$P_1 = P + P \cdot r = P(1+r)$
2	$P_2 = P_1 + P_1 r = P_1(1+r) = P(1+r)(1+r) = P(1+r)^2$
3	$P_3 = P_2 + P_2 r = P_2(1+r) =$ $P_3 = P(1+r)^3$
$\vdots$	$\vdots$
t	$P_t = P(1+r)^t$ if compounded once a year

12% per yr compounded 4 times a year.

If the annual interest rate is r and if interest is compounded n times per year, then in each time period the interst rate is $\frac{r}{n}$, and there are $n \cdot t$ time periods in t years. This leads to the following formula for the amount after t years.

memorize *

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

ex. A sum of \$1000 is invested at an interest rate of 12% per year. Find the amounts in the account after 3 years if interest is compounded annually, semiannually, quarterly, monthly, and daily.

$$A = P\left(1 + \frac{r}{n}\right)^{nt}$$

Compounding	n	Amount after 3 years
Annual	$n=1$	$A = \$1000\left(1+\frac{.12}{1}\right)^{1\cdot 3} \approx \$1404.9$
Semiannual	$n=2$	$A = \$1000\left(1+\frac{.12}{2}\right)^{2\cdot 3} \approx \$1418.5$
Quarterly	$n=4$	$A \approx \$1425.76$
Monthly	$n=12$	$A \approx \$1430.76$
Daily	$n=365$	$A \approx \$1433.24$

NOTE: The interest paid increases as the number of compounding per year n increases. What happen when $n \to \infty$?

Continuous Compound Interest

Let $m = n/r$,

$$A = P\left(1+\frac{r}{n}\right)^{nt} = P\left(1+\frac{r}{mr}\right)^{mrt} = P\left(1+\frac{1}{m}\right)^{mrt}$$
$$= P\left[\left(1+\frac{1}{m}\right)^{m}\right]^{rt}$$

For continuous compounding, the amount A in an account with principal P and annual interest rate r after t years is

$$A = Pe^{rt}$$

* compound interest formula

<u>ex.</u> Find the amount after 3 years if \$1000 is invested at an interest rate of 12% per year, compounded continuously.

$$A = 1000e^{.12(3)} = 1000e^{.36}$$

NOTE: For a given principal, interest rate and time, continuous compounding will always yield a larger amount than compounding n times a year.

ex. A radioactive substance decays in such a way that the amount of mass remaining after t years is given by the function

$$m(t) = 20\left(\frac{1}{2}\right)^{\frac{t}{10}}$$

where $m(t)$ is measured in kilograms.

1) Find the mass at time $t = 0$.

$m(0) = 20\left(\frac{1}{2}\right)^{0/10} = 20\left(\frac{1}{2}\right)^{0} = 20 \cdot 1 = 20\text{kg}$

2) How much of the mass remains after 30 years?

$m(30) = 20\left(\frac{1}{2}\right)^{\frac{30}{10}} = 20\left(\frac{1}{2}\right)^{3} = \frac{20}{8}$

$= \frac{5}{2}\text{kg}$

Practice.

1) Solve for x: $9^{-x} - 2 \cdot 3^{1-x} = 27$

2) Solve for x: $10^x - 5^2 \cdot 2^x - 2^2 \cdot 5^x + 100 = 0$

3) Solve for x: $14^x - 7 \cdot 2^x - 4 \cdot 7^x + 28 = 0$

4) Given $2^x + 2^{-x} = 5$. Find $4^x + 4^{-x}$.

Answer. 1) $x = -2$ 2) $x = 2$ 3) $x = 1, 2$ 4) 23